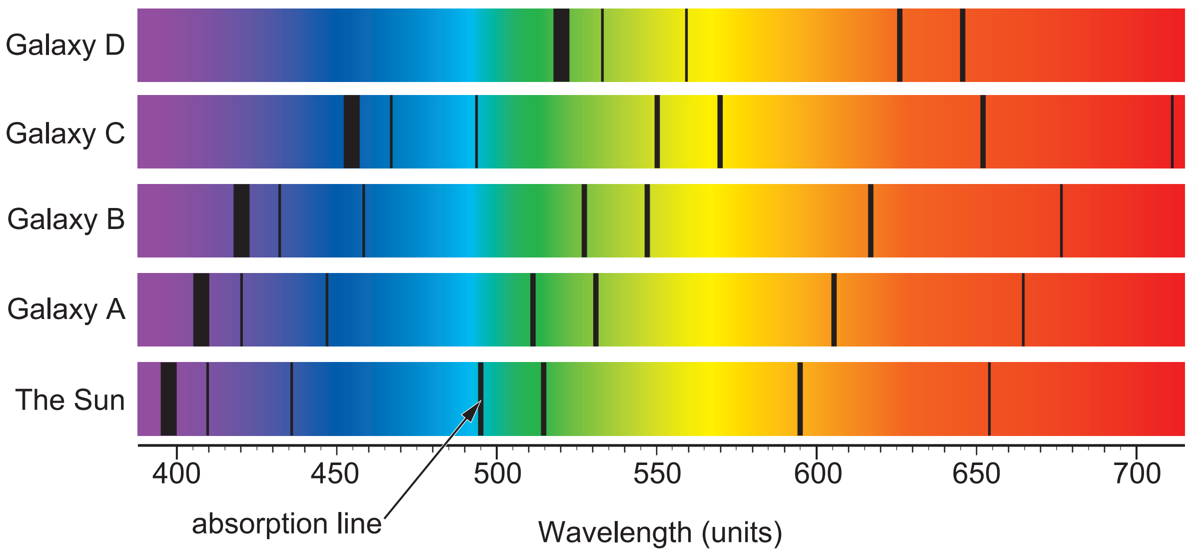
Examiner only

4. The diagram below shows the absorption spectra of 4 different galaxies compared to part of the Sun's absorption spectrum.



(a) (i) One absorption line in the spectrum from the Sun is labelled.
The wavelength of the same absorption line from each galaxy is given in the table below.
Complete the table with the estimated wavelength of that absorption line in galaxy C. [1]

Star or galaxy	Wavelength (units)
The Sun	495
Galaxy A	512
Galaxy B	528
Galaxy C	....
Galaxy D	626

(ii) The absorption lines in the galaxies are red shifted.
State which galaxy is furthest from the Sun.
Explain how you know. [2]

....

....

....

(iii) State what the red shift of these lines tells us about the Universe. [1]

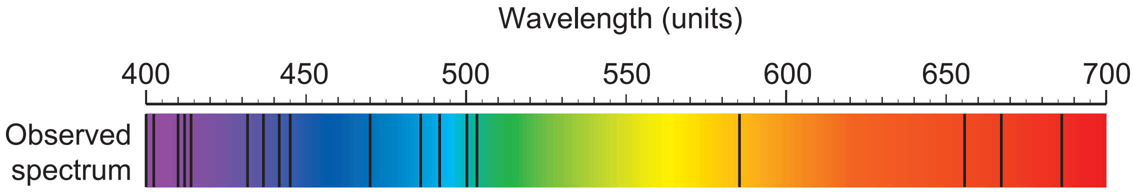
....

Examiner only

(b) Absorption lines can tell us which elements are present in a star.
The table shows the wavelengths of one absorption line from some common elements.

Element	Wavelength of absorption line (units)
helium	587
calcium	430
magnesium	516
iron	527

The observed spectrum from a star is shown below.



Rhodri states that the star contains helium and iron.
Use the information from the table and the observed spectrum to explain whether you agree. [2]

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